Essays

|!|BCEEB0030000001|!| ★

Some cells produce and secrete proteins (e.g. enzymes). Give an account of how organelles work together for the production and secretion of proteins. (10 marks)

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- The nucleus commands (1)
  ribosomes to synthesize proteins. (1)
- Many ribosomes are attached to endoplasmic reticulum which forms an extensive network spanning across the cell. (1)
- The proteins synthesized by ribosomes are transported by the endoplasmic reticulum. (1)
- Vesicles containing proteins are formed and they migrate towards the cell membrane. (1)
- The vesicles fuse with the cell membrane. (1)
- The proteins inside the vesicles are released out of the cell by exocytosis. (1)

Communication (Max. 3)

(10 marks)

##

Skills assessed	Marks
Understanding basic concepts	7
Communication	3

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|!|BCEEB0030000002|!| ★

Bacteria are unicellular organisms. Each bacterial cell must carry out all of the vital processes required for life. Explain which sub-cellular structures of a bacterial cell carry out these vital processes. (11 marks)

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- The cell membrane allows the absorption of nutrients from the environment. (1)
- Some bacteria contain chlorophyll which allows photosynthesis to occur to synthesize carbohydrates. (1)
- The cell membrane also controls the exit of substances from the cell, e.g. metabolic wastes. (1)
- The cytoplasm contains enzymes which catalyse various metabolic reactions. (1)
- DNA controls all activities of the cell, such as protein synthesis and reproduction. (1)
- Ribosomes synthesize proteins based on instructions from DNA. (1)
- Flagella can be used for locomotion/movement of the bacterial cell. (1)
- The cell wall protects the cell from damage / prevents the cell from bursting. (1)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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